

Uncertainty-Aware Contextual Recognition of Sign Language Sequences

The objective of this project is to design and implement an AI-based sign language recognition system that can recognize sign languages and translate them into text while taking into consideration contextual relationships and uncertainties. Most of the sign language recognition systems available in the literature focus on isolated sign classification, where each sign is considered in isolation from the others, or focus on sign language translation, where large amounts of data are required. The isolated classification systems may not be able to capture any relationships between signs, while translation systems are not feasible in terms of a course project. This project addresses a well-defined intermediate problem: recognizing sign sequences under uncertainty by incorporating contextual information, without attempting unrestricted natural-language translation.

The proposed system will combine computer vision, deep learning, and probabilistic sequence modelling. A convolutional neural network (CNN) will be used to extract visual features from hand gesture images or video frames and produce probabilistic predictions over a fixed vocabulary of signs. As a first baseline, a frame-based classifier will be developed, wherein predictions are carried out on a frame-wise or gesture instance-wise basis. As an extension of this baseline, a context-aware sequence model, like a Hidden Markov Model (HMM) or a neural sequence model, will be used to capture dependencies between consecutive signs. This will enable the system to make predictions based on contextual consistency rather than just visual evidence. The emphasis of the project is not on proposing a new learning algorithm, but on analysing how uncertainty-aware contextual modelling improves recognition quality compared to frame-based approaches.

The project will be implemented in Python, using OpenCV for visual preprocessing and PyTorch for neural network training and evaluation. Publicly available sign language datasets containing isolated signs will be used. To enable controlled experimentation with contextual modelling, sign sequences will be constructed from these isolated samples, allowing systematic comparison between frame-based and context-aware recognition. The evaluation would be frame accuracy, sequence accuracy, and top-k prediction accuracy, as well as a qualitative evaluation to show how uncertainty is resolved with contextual inference.

The design would also be informed by previous research on sign language recognition, as well as probabilistic models and sequence inference methods. Past research on sign language recognition with methods such as CNN classifiers, CNN-HMM, and CNN-RNN have shown good results for such methods. However, the research was conducted on final accuracy or larger benchmarks. This project, on the other hand, will focus on the evaluation framework and the importance of the role of uncertainty and context in sign language recognition rather than an end-to-end solution.

The expected outcome of the project would be a sign language recognition system that is extensible and showcases the benefits that are associated with the improvement in model robustness as a result of contextual modelling. In addition to the actual implementation of the project, the project would also be used as an opportunity to explore possibilities for extending the project, which could include making the model work as a reusable API, making the model work as part of a simple application interface, or making the model work as part of a larger multimedia application.

The project would be used as an opportunity to demonstrate concepts that are relevant and integral to the understanding of AI concepts that are covered in the course, and would include machine learning, neural networks, probabilistic reasoning, and ethics in terms of accessibility and inclusivity.